

PHI2394 • exam 1 • the map

1 Plato 428 to 348 BCE

What is actually real, and how would we find out?

- The **Forms** are perfect originals in a realm outside this one. Everything visible is a copy, so the senses show shadows.
- **Dialectic**: question and answer until opinion is sorted from knowledge.
- Education is **turning the soul around** to face the light. The eyes work; they point the wrong way.
- **Kalliopolis**: the beautiful city, run by philosopher kings.

↓ His own student keeps the word form and moves it out of that other realm.

2 Aristotle 384 to 322 BCE

What makes things move, and where do they come from?

- Form is **in the thing**, which also has a **telos**, an end it grows toward. An acorn heads for oak.
- Because form is here, you can cut animals open and learn something. The first biologist.
- Against Plato's separate realm. Attacked later by Bacon over *telos*, and by Copernicus over his cosmos.

↓ His cosmos then holds for two thousand years, so the next break is a large one.

3 The Middle Ages read by Lynn White

Why did Western Europe build the technology it built?

- The **religious picture drives the technology**. God is an architect, man is made in his image and told to rule the world.
- Christian history runs one way toward a goal, so there is no time to lose and **manual work becomes worship**.
- By 1300, **Deism**: God as a clockmaker who set it going and stepped back, so studying the mechanism is pious.

↓ Now the Greek picture starts to come apart, first in its content.

4 Copernicus 1473 to 1543

How do the planets actually move?

- The sun stands still and the earth moves around it.
- **Her framing**: he put the sun at the centre so that **circular motion** would work. He was keeping an old Greek commitment, with no new data behind it.
- Against Aristotle's earth-centred cosmos, which was also the Church's.

↓ The content has broken. Next the method breaks.

5 Bacon 1561 to 1626

How should anyone find anything out?

- **Empiricism**: the senses are where knowledge starts. And you make nature answer, by experiment.
- **Knowledge is power**, over the physical world, offered as a promise.
- **Ant** the alchemists, pile up and do nothing. **Spider** the Aristotelians, spin systems out of themselves. **Bee**, gathers then digests.
- Against *telos*: if a thing carries its purpose, you wait to receive it. He wants you to go and take it.

↓ Not everyone is pleased about the new confidence in progress.

6 Rousseau 1712 to 1778

Has progress in the sciences and arts made people better?

- **No**. Learning advanced. Morals did not.
- A society is measured by the **virtue** of its people. Great learning with no virtue is still barbarous.
- Virtue comes from **education**, on the Greek model. So that is the only route to real progress.

↓ Meanwhile institutions are being designed by arithmetic.

7 Bentham 1748 to 1832

How should an institution be arranged?

- **Utilitarianism**: the greatest happiness of the greatest number. What he was famous for in his lifetime.
- Lifelong **prison reform**. The **panopticon** was his humane replacement for hanging and filthy gaols, and was never built in Britain.
- Ring of cells, central tower, two windows per cell so the occupant shows as a silhouette.

↓ Bacon's confidence in method hardens into a doctrine about knowledge itself.

8 Positivism Comte, the Vienna Circle

What counts as real knowledge?

- **Only what can be observed and measured**. Anything else is not knowledge.
- So research must be **non-contextual**: the result cannot depend on who ran it, or where, or when.
- **Standardised**: anyone following the same procedure gets the same result.
- **Analytic**: isolate one variable, hold the rest still.

↓ Kuhn's target. A paradigm is a context, so the first of the three goes.

9 Kuhn 1922 to 1996

Does science accumulate steadily toward the truth?

- **No**. Long stable stretches, broken by revisionary breaks. Against positivism and against steady progress.
- **Paradigm**: the shared picture a field works from, with its instruments, standards and assumptions. Newton, Lavoisier, Mendel.
- **Normal science**: puzzles where the final picture is agreed and every piece leads toward it.
- **Anomaly**: a result that will not fit.
- **Revolution**: the picture gets replaced and the anomaly is explained.
- **Incommensurability**: no shared unit for measuring old against new.
- Rejects **Whig history**. A revolution is **not progress**: it builds and destroys at once.
- **Her conclusion**: science is ad hoc and disunified and works anyway.

↓ If the shape of a science can shift, the next question is who benefits from the shape it has.

10 Foucault 1926 to 1984

What are modern institutions doing to people?

- **History of Madness**, 1961: psychiatry's claimed neutrality covers for controlling people who break social rules. Homosexuality, hysteria.
- *Discipline and Punish*, 1975: a **genealogy**, how a practice came to be without assuming it improved.
- **Three techniques**: hierarchical observation, normalizing judgment, the examination. Watch, rank, record.
- **Normalization** is the goal. Her examples: standardized education, medical practice, industrial production.
- **Power-knowledge**: Bacon's equation kept, the power now over people, the verdict reversed to an accusation.
- **Docile bodies**: broken into segments, analysed, recomposed for effect.

↓ Kuhn says the content could have gone otherwise. Foucault says knowing has a politics.

11 Social construction Sismondo, Hacking 1999

In what sense is a scientific fact made?

- **Hacking**: any theory is not the only one that could have been established, so success is no evidence for it.
- Institutions are real because **enough people act as though they exist**. Politeness: no law, no building, constrains everybody.
- The test is **causal power**. Gender is real because treating people as gendered produces gendered people.
- **Kinds**, meaning categories like gold or ADHD.
- **Realist**: the world comes divided and we find the lines. **Nominalist**: we draw them, so they live in our language. Test case ADHD.

Who turns up twice

Bacon three times **Week 2**: the founder.

Empiricism, experiment, knowledge is power as a promise about nature. **Week 3**: inside Foucault, the same three words become power-knowledge, power over people, the verdict reversed. **Week 4**: a failed prediction. He expected method to level intellectual ability; ten per cent of authors produce half of all papers.

Aristotle three times **In his own right**: form in the thing, plus *telos*. **As Bacon's target**: *telos* is precisely what Bacon objects to. **As Copernicus's target**, and as Kuhn's contrast for foundationalism, since *telos* pulls from ahead and a foundation pushes from below.

Plato twice **Week 2**: the cave, and education as turning around. **Week 3**: beside Foucault, because his panoptic utopia is meant to recall Plato, the *Republic*, the Kalliopolis. One ordered city described twice, once admiringly and once as a warning.

General words	
Ad hoc	Put together for the one case in hand, with no general rule behind it. Science is ad hoc and disunified and works anyway. Wittgenstein's games: solitaire has no competition, snakes and ladders no skill, ring-a-ring-a-roses no winner. No one feature is in every game, only overlaps, which he calls a family resemblance . So there is no scientific method, singular.
Contingent	It could have been otherwise, and nothing made it necessary. Contingently real means a thing is genuinely real and could still have been built differently. Week 4 says that of knowledge, methods, disciplinary boundaries and styles of work.
Metaphysics	The study of what exists, and of what things fundamentally are. The name is a filing accident. Compilers put Aristotle's book after his <i>Physics</i> .
Dualism	A split of one thing into two. Schadewaldt's complaint is that modern thought split nature from spirit, thought from extension, and subject from object, where the Greeks had held them together.
The Greeks	
Homer	eighth century BCE The earliest writer the course reaches back to. The <i>Odyssey</i> relies on a cosmos with an order and a place for everything in it. That assumption is the ground everybody later argues on.
Heraclitus	sixth century BCE Nature is in harmony. Sound thinking is man's greatest power, and his highest aptitude is being able to say what is and fashion creatively by hearkening to nature. Harkening means listening, so the quotation assumes there is an order out there to listen to.
Aristotle	384 to 322 BCE Born in Stagira, never an Athenian citizen, studied under Plato, tutored Alexander the Great. Called the first biologist. The non-citizenship is why he had to leave Athens at the end of his life.
Geocentrism	Aristotle The earth at the centre, everything else turning round it. Falls out of <i>telos</i> : earth's natural place is the centre, which is why a rock falls, so the earth sits at the centre too. Patient observation confirmed it daily for two thousand years.
The dialogues	Plato Twenty-six of them, and Plato never speaks in his own voice. Socrates is almost always the speaker. He was a real person, and the Socrates in the dialogues is Plato's. So no line can be quoted as Plato's own opinion. The one set here is the <i>Republic</i> , and the cave is in book seven.
The Academy	Plato His school in Athens, where the English word academy comes from. Aristocratic women are known to have attended it.
The cave	Plato Prisoners chained facing a wall, taking the shadows for the real things. Out of the sensible world (what the senses reach) up to the intelligible (what only thought reaches), which Plato calls the journey of the soul. Being freed is painful and has to be compelled. The returner is laughed at; whoever unchains the others would be killed. It asks how we know what is true and what method to use, so it is a text about method.
physis	Schadewaldt Nature, meaning the source everything grows out of. It carries movement inside it, which is why the Greeks used it to answer what makes things move.
techne	Schadewaldt Skill at making or doing something. A person has to be involved, so it is never a part of nature on its own.
episteme	Schadewaldt Knowledge of what never changes. Ranked above techne, which works on things still in the middle of becoming.
empeiria	Schadewaldt Plain experience: you can do the thing and cannot say why it works.
atechnia	Schadewaldt Making something confidently out of a false idea of how it works. His word for a blunder. It can be elaborate and productive and still be one.
telos	Aristotle The end a thing is growing toward. An acorn is heading for oak. To know a thing is to know its end. The root of the English <i>teleology</i> .
Form	Plato and Aristotle What makes a thing the kind of thing it is. Plato puts the Forms in a realm outside this one and makes everything here a copy. Aristotle puts form inside the thing, together with its end. That is why Aristotle could do biology and Plato could not.
Dialectic	Plato Question and answer until opinion has been sorted out from knowledge.
Kalliopolis	Plato His beautiful city, run by philosopher kings, where the right education is what holds it together. She wrote on her own slide that Foucault's perfectly governed city should make you think of it.

Philosopher kings	Plato The rulers of his ideal city, described as awake while everyone else dreams. Their education needs geometry and astronomy, because mathematics is his route to what you cannot touch.
Method and knowledge	
Empiricism	Bacon The view that the senses are where knowledge starts. From the Greek <i>empeiria</i> , experience. Its opposite is the view that reason alone gets you there, which is Plato.
Induction	Reasoning up from the cases you have seen to a law covering all of them. A thousand swans, every one white, so all swans are white. The law covers every swan there is and you looked at a thousand, so the cases never make it safe. Europeans held that one until 1697, when a Dutch expedition found black swans in Western Australia.
Foundationalism	Every claim rests on a claim below it, down to a bottom layer needing nothing under it. The kettle boils at 100 because water does, because every measurement came out that way, and there the chain stops. The bottom layer is laid by induction , so it is the least secure part. Traced <i>down</i> through reasons, never <i>back</i> through time, which is Whig history.
Knowledge is power	Bacon Know how a thing works and you can make it do what you want. He means power over the physical world, and he means it as a promise. Foucault later keeps the equation and turns it into an accusation.
Ant, spider, bee	Bacon Three ways of working, two of them wrong. The alchemists are ants: they pile material up and do nothing with it. The Aristotelians are spiders: they spin systems out of themselves with no material coming in. The proper natural philosopher is a bee, who gathers and then digests. <i>Novum Organum</i> I.95.
Lynn White Jr.	Religious worldviews decide which technologies get built. God is an architect, man is made in his image and told to rule the world, and Christian history runs one way toward a goal where other religions had it circling. So there is no time to lose and manual work becomes worship. Written against the view that economic need drives technology.
Deism	Lynn White God as a clockmaker who set the world going and stepped back. Developing around 1300. Once the world is a mechanism, taking it apart to see how it runs becomes a pious act.
Comte's three states	Explanation passes through three stages over history: theological, then metaphysical, then positive. Gods explain things, then abstract essences and forces explain them, then only what can be observed and measured counts. The third stage is where the word positivism comes from. Note that this is a progress story, which puts Comte on the opposite side from Kuhn.
The Vienna Circle	1920s and 1930s Philosophers and scientists holding that a statement means something only if it can be checked against experience. Logical positivism, with Moritz Schlick at the centre. The hardest and most organised version of the doctrine Kuhn's book breaks with.
Galileo	1564 to 1642 Found by looking that the planets revolve around the sun. The counter-example to Aristotle's patient observer waiting for the world to reveal its purposes. He went and got the answer.
Positivism	Only what can be observed and measured counts as knowledge. So research has to be non-contextual (the result cannot depend on who ran it, or where, or when), standardised (anyone following the same procedure gets the same result), and analytic (isolate one variable, hold the rest still). Kuhn's target : a paradigm is a context, so the first of the three goes.
Realism	There is one way the world is, and we can find it out and describe it accurately. Truths depend mostly on the natural world and only slightly on the people stating them.
Kuhn	
Normal science	Solving puzzles where the final picture is already agreed, and every piece you place leads toward it. A puzzle has a guaranteed answer, so failing one counts against you. The picture stays where it is. That asymmetry is what lets a field carry on through years of bad results.
Paradigm	The shared picture a field works from, with the instruments, standards and assumptions that come with it. Her examples: Newton's mechanics, Lavoisier's chemistry, Mendel's genetics. Note how wide it is, because that width is what makes incommensurability possible.
Disciplinary matrix	Kuhn's later and more careful word for a paradigm.
Anomaly	A result the field cannot make fit, however hard it tries. They pile up, and a pile of them is what eventually breaks a paradigm.

Revolution	The shared picture gets replaced, and the result that would not fit is finally explained. Not progress, because it builds and destroys at the same time, so there is no single direction left to measure.
Incommensurability	No shared unit for measuring two paradigms against each other. Mass : fixed in Newton, changes with velocity in Einstein, so "mass is conserved" is true in one and false in the other. Three reasons : meanings of terms change, indoctrination, historical context changes. Counter-example : particle detectors survived a theory change unchanged.
Whig history	Telling the past as a line of steps leading up to us. Kuhn's objection is logical: events have to follow their causes, so a sequence cannot be explained by the fact that it ended up here.
Copernicus	1473 to 1543 He put the sun at the centre and set the earth moving. Her framing, and she called it the important one: he needed the sun at the centre for circular motion to work. He was keeping an old Greek commitment to perfect circles, with no new data behind the move.
Foucault	
Panopticon	Bentham A ring of cells round a central tower, where the inmate is always visible and can never check whether anyone is watching. Two windows per cell, so light crosses it and the occupant shows as a silhouette. The architecture does the watching, which is why one person can replace many.
Bentham's three words	Visibility, unverifiability, and external power becoming non-corporeal. Non-corporeal means the power stops needing a body in the tower. Nobody has to be up there, so it costs almost nothing to run.
Utilitarianism	Bentham The right action is whichever produces the greatest happiness for the greatest number. What he was actually famous for in his lifetime, ahead of the panopticon. He also spent much of his life on prison reform.
Three techniques of control	Hierarchical observation, normalizing judgment, and the examination. Watch, rank, record. Hierarchical observation is watching arranged in ranks. Normalizing judgment measures you against a norm. The examination is where the two meet: it watches you and writes a verdict about you in one act.
Power-knowledge	Knowing someone and controlling them turn out to be a single act. His gloss: the development of force and the establishment of truth. A file written about a prisoner becomes the truth about him and decides whether he goes home. Built on Bacon's knowledge is power, with the verdict reversed.
Normalization	A norm gets set and everyone is measured against it, so people sort into normal and abnormal. She called it the goal of disciplinary power. Her examples are all places nobody is being punished: standardized education, medical practice, industrial production.
Docile bodies	Bodies made useful and easier to control by discipline. Prisoners, soldiers, workers, schoolchildren. The body is broken into segments, analysed in detail, and recomposed for maximum effect.
Genealogy	Studying how a practice came to be, without assuming it was an improvement. The method of <i>Discipline and Punish</i> , 1975. His puzzle: punishment really did get gentler, so what is the gentler method doing that the old one could not?
The plague town	How <i>Panopticism</i> opens: the plague is met with Order. Sorting the disease, separating bodies, sorting evil, which increases fear and death. Its political dream is strict division and regulation reaching into the smallest details of daily life.
History of Madness	1961 Psychiatry's claimed neutrality covers for controlling people who break social rules. Her examples: homosexuality and hysteria. What was presented as objective discovery came out of questionable social and ethical commitments.